Module 13)>>5.Looping (For, While)

- **Introduction to for and while loops.**

Loops allow you to execute a block of code repeatedly until a condition is met. Python has two main types of loops:

1. for loops → Iterate over a sequence (lists, strings, ranges, etc.)
2. while loops → Repeat as long as a condition is True

1 For Loop :

```
for item in sequence:
    # Code to execute for each item
```

1 While Loop:

```
while condition:
    # Code to run while condition is True
```

- **How loops work in Python**

Loops in Python allow you to repeat a block of code multiple times, making them essential for automating repetitive tasks. Python has two main types of loops:

1. For loops (iterate over sequences)

2. while loops (repeat while a condition is true)

1. The for Loop:

How It Works

- A for loop iterates over an iterable (lists, strings, tuples, dictionaries, sets, etc.).
- Python secretly uses iterators (__iter__() and __next__() methods).

Key Features

✓ Works with any iterable (lists, strings, ranges, dictionaries)
✓ Automatically handles iteration logic
✓ Memory efficient (generates items one-by-one)

2. **The while Loop:**

How It Works

- Keeps executing as long as the condition is True.
- No automatic iteration – you must update the condition manually.

Key Features

✓ Flexible (can loop until any condition changes)
✓ Risk of infinite loops if condition never becomes False
✓ Used for unknown iteration counts (e.g., user input)

- **Using loops with collections (lists, tuples, etc.).**

1. Looping Through Lists

Lists are ordered and mutable, making them ideal for iteration.

Basic for Loop:

```
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)
```

2. Looping Through Tuples

Tuples are immutable, but looping works the same as lists.

```
colors = ("red", "green", "blue")
for color in colors:
    print(color)
```

3. Looping Through Dictionaries

Dictionaries store key-value pairs, and you can loop through:

- Keys (default behavior)
- Values
- Items (key-value pairs)

```
person = {"name": "Alice", "age": 25, "city": "New York"}
for key in person:  # Same as `for key in person.keys()`
    print(key)
```

4. Looping Through Sets

Sets are unordered collections of unique elements.

```
unique_numbers = {1, 2, 3, 3, 4}  # Duplicates removed
for num in unique_numbers:
    print(num)
```